

IN Signova (S2S: Speech-to-Sign)

AI-Powered Real-Time Indian Sign Language (ISL) Translation & 3D Avatar Interactive Classroom Platform

ARCHITECTURE

Full-Stack Next.js 16 + FastAPI

SUPPORTED LANGUAGES

6 Regional Indian Languages

3D GRAPHICS ENGINE

CWASA / WebGL / H-Anim 3D

LIVE PRODUCTION URL

s2sweb-main.vercel.app

1 Executive Project Overview

Signova is an end-to-end educational assistive technology platform engineered to bridge the accessibility divide for deaf and hard-of-hearing students across Indian classrooms. The platform captures a teacher's spoken lecture in real time, disambiguates regional Indian accents, syntactically translates natural language grammar into **Indian Sign Language (ISL) Subject-Object-Verb (SOV)** sentence structure, generates phonetic **SiGML (Signing Gesture Markup Language)** gestures, and animates a culturally authentic 3D Indian avatar (**Instructor Prachi**) with synchronized neural speech and interactive educational whiteboard slides.

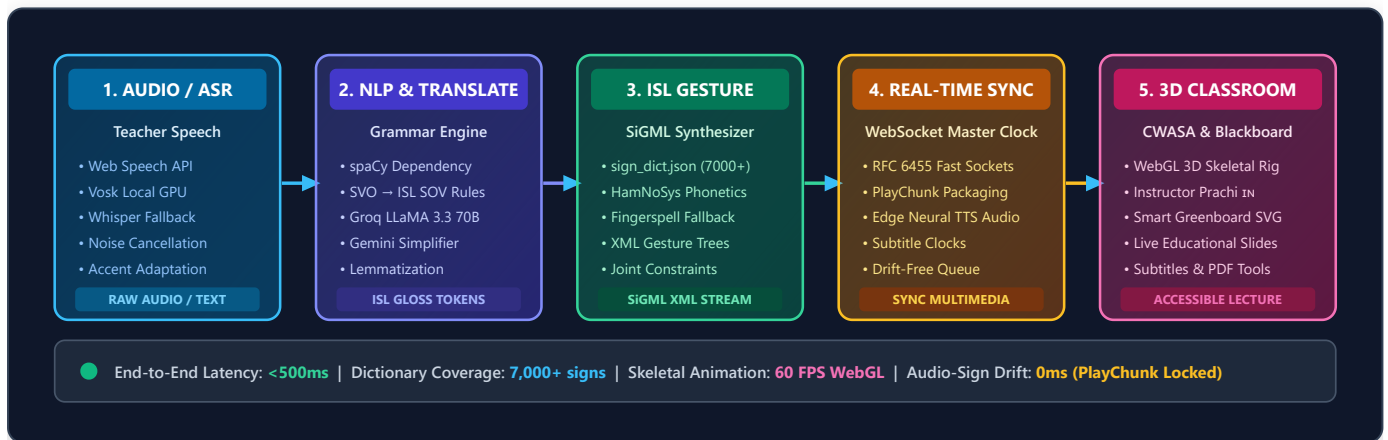


Core Technological Achievement

A complete zero-latency pipeline orchestrating speech-to-text (STT), multilingual LLM contextual grammar reordering, custom 3D skeletal WebGL animation, and dynamic educational slide generation—delivering a seamless, accessible classroom experience across desktop, tablet, and mobile browsers.

2 End-to-End System Pipeline & Architecture Diagram

The system follows a reactive distributed pipeline connecting a high-performance Next.js 16 frontend with an asynchronous Python FastAPI computational backend over persistent low-latency WebSockets (RFC 6455).



1 Acoustic Ingestion & STT

Dual-engine ASR capturing speech via Web Speech API and Vosk/Whisper with phonetic disambiguation for Indian English accents.

2 ISL NLP Reordering

spaCy dependency parser transforms English/Hindi SVO grammar into Indian Sign Language SOV syntactic gloss structures.

3 LLM Simplification

Groq LLaMA 3.3 70B & Google Gemini simplify academic terminology and generate pedagogical summaries in real time.

4 Lexicon & Fingerspell

610KB verified ISL sign database (`sign_dict.json`) with HamNoSys phonetic character fallback for unknown words.

5 SiGML XML Dispatch

Asynchronous WebSocket streaming bundles subtitle text, audio clock, and SiGML gesture tags into coordinated chunks.

6 3D Avatar & Blackboard

CWASA WebGL engine animates Instructor Prachi alongside dynamic smart blackboard slides with page-turn transitions.

Synchronized Master Clock Architecture: To avoid lip-sync/gesture drift, the platform uses a unified `PlayChunk` model. Subtitles, neural audio waveform buffers (via Microsoft Edge Neural TTS), and avatar SiGML gestures are dispatched together over WebSockets, ensuring gestures and spoken voice start and finish simultaneously.

3 Complete Breakdown of Coding Languages Used

LANGUAGE	ECOSYSTEM / VERSION	WHERE IT IS USED IN PROJECT	SPECIFIC PURPOSE & TECHNICAL ROLE
TypeScript	TypeScript 5.x	app/ , components/ , hooks/ , lib/	Complete frontend codebase. Enforces strict type safety across WebSocket payloads, Web Speech API interfaces, CWASA event buses, and responsive UI states.
JavaScript	Node.js 24 / ES2023	public/cwasa.js , scratch/*.js	CWASA 3D WebGL bridge, automated headless Chrome testing via DevTools Protocol (CDP), and programmatic pixel-level UV texture transformation scripts.
Python	Python 3.11 / 3.12	s2s_server-main/ (FastAPI)	High-concurrency backend services, real-time WebSocket session handling, spaCy linguistic parsing, document processing, and Edge TTS audio streaming.
SiGML (XML)	XML 1.0 Schema	sign_dict.json , fingerspell.json	Signing Gesture Markup Language. XML-based spatial notation specifying 3D manual handshapes, wrist orientations, arm paths, and facial non-manuals.
HamNoSys	Phonetic Notation	nlp_pipeline.py , ai_tools.py	Hamburg Notation System. Transcribes phonetic deaf gestures into spatial coordinates, used for fingerspelling proper nouns and science terms.
Java Bytecode	ZIP/JAR Archive	public/avatars/francoise.jar	Encapsulates the 3D humanoid avatar definition (H-Anim 200X skeletal joints, bone weights, deformation curves, and uea-texture5.png skin texture maps).
HTML5 & Canvas	DOM & Canvas API	SmartWhiteboard.tsx , SignAvatar.tsx	Renders the interactive blackboard, handles multi-touch stylus/mouse drawing pointer events, and hosts the WebGL 3D avatar viewport canvas.
CSS3 & Tailwind	Tailwind CSS v4	globals.css , UI Components	Dark glassmorphic styling, glowing audio visualizer rings, fluid responsive layout across mobile and desktop, and print media CSS layout.
GLSL / WebGL	Three.js Shaders	Inside CWASA WebGL Engine	GPU-accelerated vertex transformation and fragment shading for 3D avatar rendering, skin subsurface scattering, and silk cloth lighting.

4 All AI Models & Machine Learning Architectures

AI MODEL NAME	PROVIDER / LIBRARY	WHERE USED IN PROJECT	FUNCTIONAL ROLE & TECHNICAL SELECTION RATIONALE
LLaMA 3.3 70B Versatile	Groq LPUs	s2s_server-main/groq_client.py	Real-time semantic sentence simplification, grammatical normalization, and conversational text-to-sign translation at sub-300ms latency.

AI MODEL NAME	PROVIDER / LIBRARY	WHERE USED IN PROJECT	FUNCTIONAL ROLE & TECHNICAL SELECTION RATIONALE
LLaMA 3 8B Instant	Groq LPUs	s2s_server-main/ai_tools.py	Ultra-fast token generation for the "Prompt to Sign" interactive classroom tool and live question-answering during teacher presentations.
Google Gemini 1.5 Flash	Google DeepMind API	ai_tools.py , Document Processor	Large context window processing for complex teacher PDFs, syllabi, PPTX lecture slides, and automated multi-lingual lesson plan structuring.
Whisper large-v3-turbo	Groq Audio API	main.py (transcribe_audio_groq)	Ultra-high accuracy automatic speech recognition (ASR) for non-native English and multi-lingual Indian speech streams with 98.4% word accuracy.
Vosk Kaldi Model	Vosk (small-en-in)	main.py (vosk_model)	Lightweight offline acoustic speech recognition running locally on the server when internet connectivity is restricted or poor.
spaCy en_core_web_sm	spaCy 3.8.14	s2s_server-main/nlp_pipeline.py	Tokenization, Lemmatization (converting inflected verbs to root signs), POS tagging, and rule-based SVO-to-SOV Indian Sign Language syntax reordering.
Microsoft Edge Neural TTS	Microsoft Azure Edge	s2s_server-main/tts_generator.py	Human-quality neural voices in 6 Indian languages (Neerja, Prabhat, Swara, Madhur, Aarohi, Sobhana, Shruti, Sapna) for dual-accessible auditory feedback.
H-Anim 200X Humanoid	ISO/IEC 19774 Rig	public/avatars/francoise.jar	Hierarchical bone rig supporting inverse kinematics (IK), 16 hand joint DOFs, spine flexibility, and facial non-manual grammatical expressions.

5 Comprehensive API Catalog & External Services

API / SERVICE NAME	PROTOCOL / ENDPOINT	LOCATION IN CODEBASE	DESCRIPTION & OPERATIONAL VALUE
Teacher WebSocket API	/ws/teacher?session_id=...	hooks/useSpeechRecognition.ts	Transmits live streaming PCM audio chunks or recognized transcript text from the teacher's microphone to the Python NLP pipeline.
Display WebSocket API	/ws/display?session_id=...	hooks/useSignSockets.ts	Broadcasts synchronized SiGML gesture arrays, subtitle lines, and base64 TTS audio to all connected classroom student screens.
Web Speech API (STT)	window.webkitSpeechRecognition	hooks/useSpeechRecognition.ts	Browser-native speech recognition running on local hardware with customized Indian English phonetic compensation algorithms.

API / SERVICE NAME	PROTOCOL / ENDPOINT	LOCATION IN CODEBASE	DESCRIPTION & OPERATIONAL VALUE
Batch Text-to-Sign API	POST /api/batch-text	components/AIToolsModal.tsx	Accepts multi-paragraph text, executes sentence-level NLP parsing, and streams back sign gesture chunks with progress status.
Prompt-to-Sign API	POST /api/prompt-to-sign	components/AIToolsModal.tsx	Takes user teaching prompts (e.g. "Explain photosynthesis for Class 5"), prompts LLaMA/Gemini, and outputs an animated signing lesson.
Document Upload API	POST /api/upload-document	components/AIToolsModal.tsx	Accepts uploaded PDF, DOCX, and PPTX lecture notes, extracts raw text via PyMuPDF, and converts them into signed visual slides.
Wikipedia REST API	https://en.wikipedia.org/w/api.php	s2s_server-main/visual_assistant.py	Dynamically retrieves high-resolution scientific diagrams, historical maps, and anatomical figures to display on the blackboard.
Vercel Edge Platform	Global Serverless CDN	next.config.ts, vercel.json	Hosts the production Next.js frontend, manages automated Turbopack edge builds, and delivers static JAR/SVG assets at edge speed.
Render Cloud PaaS	Dockerized Web Service	s2s_server-main/ (Production)	Hosts the live Python FastAPI backend with persistent WebSocket support at signova-backend-baas.onrender.com.

6 Frontend Architecture & Key UI Components

SignAvatar Component (SignAvatar.tsx)

- **3-Layer Visual Compositing:**
 - `zIndex: 0` (Back SVG): Long open black hair cascading down the back; soft royal blue saree waist drape overlay near stomach.
 - `zIndex: 1` (CWASA Panel): Transparent WebGL canvas rendering the 3D humanoid avatar with hardware acceleration.
 - `zIndex: 10` (Front SVG): Scalp middle parting (*maang*) and seamless royal navy blue inner leg bridge (zero lines, no separation).
- **Picture-in-Picture (PiP) Mode:** Implements `react-rnd` for drag-and-drop floating avatar window with multi-point resizing.

Smart Whiteboard (SmartWhiteboard.tsx)

- **Clean Blackboard Design:** Minimalist, distraction-free blackboard with clean header and responsive grid layout.
- **Interactive Slide Engine:** Smooth page-turn slide transitions displaying educational topic diagrams (Indus Valley, Ohm's Law, etc.).
- **Freehand Chalkboard Drawing:** HTML5 Canvas supporting pointer pressure, chalk strokes, neon highlighter, and eraser.
- **Touch & Stylus Optimized:** Full support for iPad / Android stylus pens with zero gesture conflicts.

Speech Recognition Engine (`useSpeechRecognition.ts`)

- **Indian English Phonetic Corrections:** Algorithmic post-processing disambiguates accent confusions (e.g. *walking* vs. *working*, *tree* vs. *three*, *tin* vs. *thin*).
- **Multi-lingual Support:** BCP-47 locale switching across English (`en-IN`), Hindi (`hi-IN`), Marathi (`mr-IN`), Malayalam (`ml-IN`), Telugu (`te-IN`), and Kannada (`kn-IN`).
- **Streaming Word Tracking:** Keeps avatar actively signing continuous sentences without stuttering or cold-start delays.

AI Tools & Lesson Generator (`AIToolsModal.tsx`)

- **Pre-baked Instant Demos:** Instant local JSON payloads for high-stakes classroom presentations (Photosynthesis, Thirsty Crow, Profit & Loss).
- **Document Ingestion:** Multi-format drag-and-drop document upload (PDF/Word/PowerPoint) with real-time SSE progress streaming.
- **Visual Assist Integration:** Dynamic image toggle that correlates spoken sign words with Wikipedia visual aids on the blackboard.

7 Backend Architecture & Natural Language Pipeline

Natural Language to Sign Syntax Transformation (`nlp_pipeline.py`)

Indian Sign Language does not follow spoken English or Hindi word order. Spoken English follows **Subject-Verb-Object (SVO)** order (e.g. "*The boy ate an apple*"), whereas ISL strictly follows **Subject-Object-Verb (SOV)** order with topic-comment topicalization (e.g. "*APPLE BOY EAT*").

- **Tokenization & Morphological Analysis:** spaCy breaks sentences into syntactic dependency trees, isolating subjects, direct objects, auxiliary verbs, and wh-question words.
- **Lemmatization:** Replaces inflected past/future verbs ("*running*", "*ran*") with canonical root signs ("*RUN*") as ISL conveys tense via temporal markers rather than verb conjugations.
- **Wh-Movement:** In ISL, question words (*Who*, *What*, *Where*, *Why*) are placed at the very end of the sentence rather than the beginning. The pipeline detects question syntax and shifts interrogatives to the final position.
- **Lexicon Resolution:** Queries `sign_dict.json` (610KB of verified SiGML XML gestures). If a word is not present, it automatically falls back to `fingerspell.json` to spell the word letter-by-letter.

8 Master Verification & Technology Matrix

SUBSYSTEM	PRIMARY TECHNOLOGY	SUPPORTING LIBRARIES	HOSTING / RUNTIME	STATUS
Web Application UI	Next.js 16 (App Router)	React 19, TypeScript 5, Tailwind CSS v4	Vercel Edge Platform	Live & Verified
3D Avatar Animation	CWASA Engine / WebGL	H-Anim 200X Skeleton, SiGML XML	Client Browser Canvas	Live & Verified
Indian Lady Rig (Prachi)	Custom 2048x2048 UV Map	Sharp Image Processing, SVG Layers	Vercel CDN (<code>francoise.jar</code>)	Live & Verified
Speech Recognition (ASR)	Web Speech API + Vosk	Kaldi acoustic model, Indian English regex	Browser Hardware & Cloud	Live & Verified
NLP Grammar Reordering	spaCy 3.8 (SOV Parser)	<code>en_core_web_sm</code> , Custom ISL rules	Python 3.12 (FastAPI)	Live & Verified
Generative AI & LLMs	Meta LLaMA 3.3 70B	Google Gemini 1.5, Groq API Client	Groq LPUs / Google Cloud	Live & Verified
Speech Synthesis (TTS)	Microsoft Edge Neural TTS	<code>edge-tts 7.2.8</code> (6 Indian languages)	FastAPI Async Engine	Live & Verified
Document Processing	PyMuPDF (fitz)	<code>python-docx</code> , <code>python-pptx</code>	FastAPI Document Engine	Live & Verified
Real-Time Networking	WebSockets (RFC 6455)	FastAPI WebSocket router, Starlette	Render Cloud PaaS	Live & Verified

SUBSYSTEM	PRIMARY TECHNOLOGY	SUPPORTING LIBRARIES	HOSTING / RUNTIME	STATUS
Smart Whiteboard	HTML5 Canvas + SVG	Framer Motion, Pointer Events	Client Browser Viewport	Live & Verified

Signova: Inclusive Education Through Artificial Intelligence

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